

High-Low Tech Ombro-Cinéma

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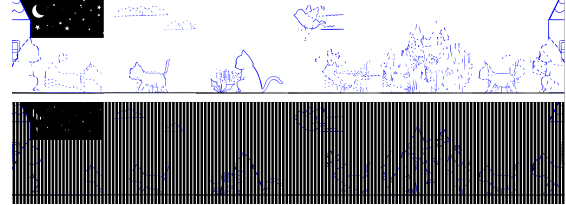


Figure 1: The High-Low Tech Ombro-Cinéma prototype (left, middle) and an example story: the interlaced image and a frame extracted with an overlay (right).

ABSTRACT

Barrier-grid animation is a primitive animation technique that utilizes an interlaced image with a transparent striped overlay. *Ombro-Cinéma*, a toy developed in 1921, utilizes this technique effectively with a scroll of graphics. By rotating an attached hand-crank, it presents a scrolling animated story, the narrative of which is often tightly associated with its mechanical structure. This paper describes *High-Low Tech Ombro-Cinéma*, a prototype device that extends Ombro-Cinéma using an embedded system. Conductive adhesive tape strips are affixed to the back side of a scroll and used to trigger various sonic events to enhance its narrativity. Such an investigation into the reproduction of an obsolete toy with digital technology may provide an interesting opportunity to review the narrative of an obsolete device and to reconsider the effects tangible interfaces have on narratives.

CCS CONCEPTS

• Applied computing → Media arts; • Human-centered computing → Interface design prototyping.

KEYWORDS

barrier-grid animation, scanimation, high-low tech, interaction design, digital arts

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1 INTRODUCTION

Barrier-grid animation is a well-known historical technique to animate images in print. It utilizes a transparent striped overlay (a barrier-grid) over an interlaced image and viewers can enjoy a tiny animation that consists of a few frames by moving the overlay. Despite its simplicity and limited capability for animation, barrier-grid animation has survived in various forms from its first appearance in the early 1900s until today.

One notable historical application of barrier-grid animation was *Ombro-Cinéma*, a type of toy that was popular in France in the 1920s[Chiou 2016, p.16]. To overcome the restriction imposed on the narrative by the limited capability for animation of the original form of barrier-grid animation, Ombro-Cinéma prints interlaced images on a scroll. By rotating the scroll with an attached hand-crank, showing viewers only a part of the entire scroll at a time, it enhances the narrativity beyond the original form of barrier-grid animation. Ombro-Cinéma can present a longer story continuously, which is not possible with the original barrier-grid animation technique.

Though Ombro-Cinéma is already considered obsolete as a display device, it may still be an interesting exemplar of interaction design as that its mechanical architecture is strongly associated with the narrative for its animated contents. We have therefore developed *High-Low Tech Ombro-Cinéma*, which extends traditional Ombro-Cinéma, using simple electric circuits and a single board computer, to explore how such narratives can be further broadened using digital technologies. While our project is still in its preliminary stages, it provides us with a good opportunity to review historical devices as an inspiration for the narrative in interaction design, not just as obsolete technologies from the past.

2 OMBRO-CINÉMA

Barrier-grid animation techniques were widely known from at least the beginning of the 20th century, and Spiegel filed a related patent in 1906 [Spiegel 1906]. It utilizes an interlaced image, which consists of a few frames to be animated, and a transparent, striped



Figure 2: An example of Ombro-Cinéma, courtesy of copy-right owner ©www.antiquetoysandgames.com.

overlay. By moving the overlay over the interlaced image, every single frame can be extracted in turn, enabling viewers to see the animated image. Figure 1 (right) shows an example of an interlaced image and a transparent overlay. Since its inception, barrier-grid animation has been repeatedly ‘re-invented’ in various forms.

Ombro-Cinéma, which was quite popular in France in the 1920s, is one of the more well-known variants of barrier-grid animation. As shown in Figure 2, it prints an interlaced image on a scroll. By rotating the attached hand-crank, a limited area of the scroll is continuously shown to viewers. Some Ombro-Cinéma products have a music box embedded within them to further enhance their cinematic experience. In Figure 2 (right), a music box is placed at the top left of the box. Notably, the narratives of stories for Ombro-Cinéma are often strongly associated with their mechanical architecture. For instance, Figure 2 shows various scenes seen in one street that are be continuously presented to viewers.

3 HIGH-LOW TECH OMBRO-CINÉMA

Such a strong association between the mechanical architecture and the narrative of a story in Ombro-Cinéma could be an interesting topic to investigate concerning interactive storytelling with a tangible interface, even in our time. Hence, we have just begun prototyping High-Low Tech Ombro-Cinéma, a device that extends Ombro-Cinéma using a high-low tech approach¹ (Figure 1).

Figure 3 shows pictures of the system design for our High-Low Tech Ombro-Cinéma device. As a preliminary experiment, we employed a simple technique that utilized conductive adhesive tape strips as digital switches and affixed them behind the scroll of the barrier-grid animation (left). These conductive tapes are detected via the GPIO of a Raspberry Pi to trigger prepared sound events associated with each scene: background music and one-shot sound effects (middle and right). Due to the ways the system is designed, High-Low Tech Ombro-Cinéma can suitably synchronize sound events to a scene. This was not possible in the original form of Ombro-Cinéma, which had an embedded music box. Thus, the major benefit of integrating an embedded system is that it enhances the expressibility of Ombro-Cinéma. It should be also noted that, unlike the original form of Ombro-Cinéma, our prototype connects both ends of the scroll to make a looped animation: the first and last scenes are connected and the story is circularly presented. Figure 1 (middle) shows a user viewing an animation with our prototype device.

¹“High-low tech is a craftmaking and design practice that encourages makers to merge high- and low-tech materials, processes, and cultures”[Kadish and Dulic 2015].

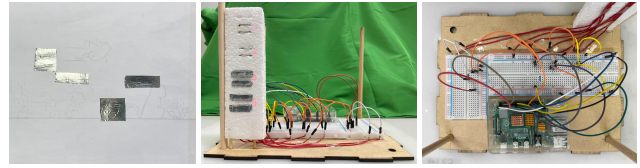


Figure 3: The High-Low Tech Ombro-Cinéma prototype.

We created a simple story so that the narrative could be associated with the system design of our prototype. As the story shown by our High-Low Tech Ombro-Cinéma is looped, we decided to base it on a ‘there and back again’ archetype. Seen in, for example, *The Lord of the Rings* by J. R. R. Tolkien, this archetype is where the main character goes out for an adventure and, after overcoming various hardships, then returns to the origin of the journey. Our target users are children so we created a short story about a kitten who spends the whole day away from his mother. He experiences various events such as an encounter with a dog, a chase with a flying bird, and so on. The story is designed to imply the mental growth that children experience (Figure 1, right).

Our project is still in its preliminary stage, but we have already found that our approach, in which an obsolete device is reproduced with digital technology to study the narrative strongly associated with its mechanical architecture, provides a good opportunity to reconsider the narrative in interactive storytelling with a tangible interface. While it is still possible to further extend expressibility through the use of other elements such as lights and haptics, the mechanical architecture would still impose a certain level of restriction on how effective the narrative can be. Such constraints would therefore be of significance as design criteria.

4 CONCLUSION AND FUTURE WORK

We developed our initial prototype of *High-Low Tech Ombro-Cinéma* intending to investigate the strong association between the narrative and the mechanical architecture in Ombro-Cinéma by reproducing it using the high-low tech approach. Regardless of its technical simplicity, the high-low tech approach was quite helpful in extending the narrative of Ombro-Cinéma, here with sound effects and musical materials, while still keeping its narrative strongly associated with its mechanical architecture. Such an investigation, reproducing an obsolete device with digital technology, could be an interesting approach to take to the relationship between narrative and tangible interfaces. We plan to extend our prototype with lights and haptics for further investigation, together with a user study involving children.

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